Vincent Zhen

(416 -897-0955) | vhzhen@uwaterloo.ca | linkedin.com/in/vincent-zhen-190885351 | github.com/vhzhen

Education

University of Waterloo — Waterloo, ON

Candidate for Bachelor of Computing and Financial Management — *September 2024 – April 2029*

- Recipient of the University of Waterloo President's Scholarship of Distinction
- Achieved Excellent Academic Standing with Term Distinction for a **92% CGPA**
- Awarded the **KPMG Scholarship** for outstanding and exemplary academic achievement

Experience

ETF Trading - Global Markets — Toronto, ON

TD Securities — *January 2026 – April 2026*

- Built a PyQt5 analytics tool to analyze cross-desk ETF flow (DMA, ATS, ETF), identifying spread inefficiencies across tens of thousands of orders and enabling targeted adjustments to quoting strategies
- Engineered a FIX protocol GUI to automate the creation and redemption of ETFs, with custom payloads for major providers including BlackRock and Vanguard, accelerating order execution **from 3–4 minutes to less than 2 seconds**
- Refactored a Visual Basic ETF parser into Python to process thousands of creation and redemption orders, performing error checking via Bloomberg API, integrating SQL database access and logging

Software Developer — Toronto, ON

Pepper — *April 2025 – August 2025*

- Expanded Pepper's Smart Attribute Library Taxonomy platform (SALT) to centralize, enrich, and standardize product data for foodservice distributors, encompassing **3M+** unique items
- Scraped **300,000+** catalogue items from major foodservice distributors using Playwright, extracting GTINs, images, allergens, and nutrient data
- Automated AI-powered website creation with Python, n8n, Snowflake, and 10Web, securing **$10,000+ ARR** in distributor contracts and advancing **10+ active leads**
- Spearheaded a large-scale image reordering project using OpenAI Vision to reorganize **1M+ images** across **400,000+** unique items in SALT, improving thumbnail image accuracy to over **95%**
- Designed and deployed **ETL** pipelines for Syndigo, Frosty Acres, and Essendant via API/SFTP integrations, ingesting **450,000+** new items into MongoDB and AWS S3 using scheduled AWS Glue runtimes

STEM Workshop Lead — Markham, ON

YRDSB — *May 2023 – June 2024*

- Co-founded an organization aimed at teaching STEM concepts with in-person interactive workshops
- Utilized Arduino and Micro:Bit kits to deliver hands-on STEM learning with real-time constructive feedback
- Hosted **4** comprehensive coding and experiment-based workshops, engaging and educating over **100** students

Projects

Robo-Advisor | *Python, NumPy, pandas, yfinance, Matplotlib*

- Designed a technical analysis algorithm to generate the most profitable portfolio from a list of given tickers
- Led a team of 3 in developing ticker filters, volatility indicators, and correlation indicators for the algorithm
- Analyzed **2+ years** of historical stock data with Pandas and NumPy to uncover correlations and visualize those insights using Matplotlib

Patient Intake Form | *Typescript, Next.js, Tailwind CSS, Supabase, GraphQL*

- Developed a production-ready patient intake platform using Next.js, GraphQL, Supabase, and Vercel Blob, including validated multi-field forms and secure file uploads.
- Built an admin dashboard with SSR, AG Grid, and optimized PostgreSQL schemas.

Frequency Filter to Combat Hearing Loss | *Python, Typescript, Matplotlib, Next.js*

- Crafted a simple yet practical GUI with Tkinter, visualizing frequency fluctuations with Matplotlib
- Received the **Best Beginner Hack award** for IgnitionHacks out of **43** teams

Technical Skills

Languages: Python, C, C++, Visual Basic, SQL, HTML/CSS, JavaScript, Racket, TypeScript, Bash
Tools: AWS S3, AWS Glue, MongoDB, Supabase, OpenAI, Gemini, Git, Next.js, Excel, Jenkins, n8n, Make
Libraries: pandas, requests, Scikit-learn, boto3, NumPy, Matplotlib, Playwright